

MOHAMED AHMED EL HARAKANI

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Education

Alexandria University

Sep. 2020 – June 2024

Bachelor of Science in Computer and Data Science (major AI), GPA:3.828

Alexandria

Coursework and Cerifications

- Algorithms and Data Structures
- Deep Learning
- Machine Learning
- Database Management
- Artificial Intelligence
- Probability and Statistics
- Linear Algebra
- Computer Vision
- Natural Language Processing
- LLM

Tensorflow Certification: [Certification](#) ([Google Developers Certification](#))

AWS Cloud Technical Essentials: [Certification](#)

Generative Deep Learning with TensorFlow: [Certification](#) Provided by DeepLearning.AI

Natural Language Processing: [Certification](#) Provided by DeepLearning.AI

Agentic AI: [Certification](#) Deep Agents with LangGraph, [Certification](#) Deep Research with LangGraph

Experience

Nokia

August 2025 – Present

AI Engineer

Cairo – Egypt

- Design and develop AI-powered agents for enterprise use cases.
- Build and optimize Retrieval-Augmented Generation (RAG) systems to enhance information retrieval and response accuracy.
- Conduct internal sessions and knowledge-sharing workshops on AI technologies and best practices.

Singularity Labs

January 2025 – March 2025

AI/LLM

Cairo – Egypt

- Worked on Sales Agent AI, improving response accuracy using RAG-based search techniques.
- Worked with LangChain and FAISS for optimized retrieval.
- Developed expertise in Retrieval-Augmented Generation (RAG) and AI Agents.
- Improved response accuracy of Sales AI using RAG and embedding-based search.
- Tested and fine-tuned LLMs (GPT-4, LLaMA) using Hugging Face and LoRA.

TechWebInnovations OÜ

September 2024 – August 2025

Junior Web Designer and Infrastructure Admin

Germany

- Design and develop custom websites for clients using platforms like WordPress, WooCommerce, and Divi.
- Create visually appealing, user-friendly websites tailored to client needs, ensuring brand consistency and optimal user experience.
- Integrate AI-driven features and tools into websites enhance functionality and client satisfaction.

Egyptian Projects Operation and Maintenance (EPROM)

August 2022 – October 2022

Database Developer Intern

Alexandria – Egypt

- The company wanted to change their database system from using only papers to being digital.
- I was responsible for making the database related to employees for sending requests to the manager (e.g., if they want to take a vacation) and the manager database that has the authority to accept or deny.
- Used SQL, PL/SQL, and Oracle Apex for this project.

Projects

Digital Twin – Network Topology Extractor

- Parses network topology diagrams from PowerPoint files, extracting nodes, connections, and link types (e.g., by color/style) using structured LLM output with Pydantic models, and exports results as YAML topology files ready for direct use in DTaaS (Digital Twin as a Service).
- Features a chatbot interface for querying the topology (e.g., “how many nodes?”, “list physical connections”) and generates an LLM-powered summary of the extracted network topology.

BMW logistics

- Generated emails using an LLM model and sent them automatically.
- The companies will reply to the emails and provide their offers and the LLM will analyse which offer suitable to the company based on specific criteria
- Implemented using Python,LangChain,Ollama,FastAPI

Hieroglyphic OCR and Translator

- Developed an OCR and translator for hieroglyphic characters. Tourists can take a picture of an Ancient Egyptian wall and receive a translation of what is written on the wall.
- Implemented using Python, Tensorflow, Keras, and the concept of transfer learning (Detectron2 model).

MonoTalk (Graduation Project)

February 2023 - June 2024

- Developed a mobile application and a web application for real time interaction with monuments in the museums and share our culture all over the world.
- Implement a CNN model for place detection and provide the right information about the place uploaded to the model (create a model from scratch using tensorflow ,then using ResNet (transfer learning concept).
- Implement a multilingual sentiment analysis model by concatenate three difference models one for detecting the comment language,one for translation and finally the sentiment analysis model(BERT model).
- Using text to speech and speech to text for the interaction with the LLM.

Skills

Technical Skills

- **Languages/Libraries:** Python, Java, Node.js, Tensorflow, C++, HTML/CSS,React, NextJs, FastAPI, JavaScript, SQL/MySQL, MongoDB, Numpy, Matplotlib, Pandas, Langchain, Langgraph, Langsmith, CrewAI
- **Developer Tools:** VS Code, Eclipse, Netbeans, Google Colab, Android Studio
- **Technologies/Frameworks:** GitHub, Docker

Languages

- Arabic (Native),French (Fluent),English (Fluent),German (Basic)